

Chapter 4

Boyle's Law

4.1 Objectives

To determine atmospheric pressure in the laboratory using the graphical method

4.2 Tools and Materials

1. U-tube contains mercury (Hg), and one of its arms is equipped with a cap
2. Ruler fixed to a stand

4.3 Theoretical Foundation

You know that we live at the bottom of an ocean of air. This air envelops the earth and is called the earth's atmosphere. The higher one goes above the earth's surface, the lower the air density becomes. The air exerts pressure on every object at the earth's surface from all directions. This pressure by air is called atmospheric pressure or external air pressure. Of course, the higher the location, the lower the atmospheric pressure, and the greatest atmospheric pressure is found at sea level. The pressure at sea level is called 1 atm or 76 cmHg . This pressure is equivalent to the vertical pressure exerted by 10 stacked sedan car hoods. Atmospheric pressure decreases by 1 mmHg for every 10 meters of elevation gain. Information about air pressure values is useful for weather forecasting, because if the air pressure in one location is significantly lower than expected, a storm will soon occur. In addition, atmospheric pressure

values are also useful for determining the altitude of a location relative to sea level, including determining the altitude of an aircraft.

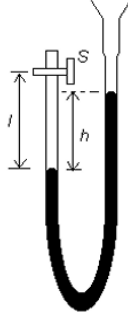


Figure 4.1: Chart of the experimental apparatus

Gases are classified into two types, namely ideal gases and real gases. Ideal gases occur in containers at low pressure. Considering an ideal gas in a closed tube with volume V , at pressure P , temperature T , number of gas moles n , and universal gas constant R , it is governed by Boyle's law

$$PV = nRT \quad (4.1)$$

If n and T are constant, then nRT equals a constant C , so that equation 4.1 becomes:

$$PV = C \quad (4.2)$$

Equation 4.2 together with the hydrostatic pressure equation can be used to determine the atmospheric pressure (external air pressure) in the laboratory P_0 , and is expressed in $cmHg$. One of the instruments that can be used to measure gas pressure is a manometer containing mercury. One side of the tube must be

isolated from external air to observe the height difference of the mercury surfaces between the left and right side. From this difference in height, we can determine the pressure difference between both tube sides. In this experiment, the left side of the tube can be isolated from the atmospheric pressure by closing the valve on that side. If the atmospheric pressure is P_0 , the density of mercury is ρ , and the acceleration due to gravity is g , then the pressure on the mercury surface in the left tube when valve is closed (see Figure 4.1) can be determined using the hydrostatic pressure equation

$$P = P_0 + \rho gh \quad (4.3)$$

P is the gas pressure in the left tube and h is the height difference between the mercury surface on the left and right sides of the tube. The volume of gas in the left side of the tube can be calculated using the following equation:

$$V = \pi r^2 l \quad (4.4)$$

V is the gas volume, r is the radius of the left tube (assumed constant), and l is the length of the air column in the left side of the tube. If equation 4.2 is substituted into equation 4.4, the following is obtained:

$$\frac{C}{P} = \pi r^2 l \quad (4.5)$$

The above equation can be simplified to:

$$P = \frac{K}{l} \quad (4.6)$$

$K = \frac{c}{\pi r^2}$ is a constant. It can be seen that the gas pressure in the left column is inversely proportional to the length of the gas column. Then, equation 4.3 and 4.6 are equated to obtain:

$$\frac{K}{l} = P_0 + \rho gh \quad (4.7)$$

And with a slight modification, the following is obtained:

$$h = \frac{K}{\rho gl} - \frac{P_0}{\rho g} \quad (4.8)$$

4.4 Experimental Methods

4.4.1 Data analysis

Equation 4.8 serves as the basic for analyzing the observational data obtained. To ensure the data exhibit a linear relationship when plotted, the y-axis is set as h (cm) and the x-axis is set as $\frac{1}{l}$ (cm⁻¹), as shown in Figure 4.2.

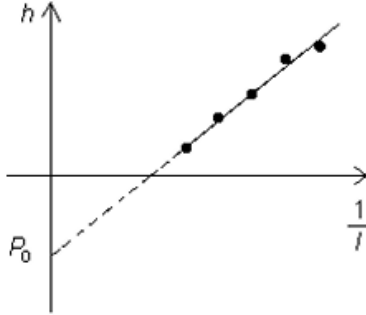


Figure 4.2: Graph of h versus $1/l$

The atmospheric pressure (P_0) is indicated by the intercept on the graph of h versus $1/l$. According to equation 4.8, the intercept (c) on the h versus $1/l$ graph is:

$$c = -\frac{P_0}{\rho g} \quad (4.9)$$

By substituting the value of c in cm and setting $\rho g = 1$ into equation 4.9, the atmospheric pressure can be obtained in units of $cmHg$. The uncertainty associated with equation 4.9 is:

$$\Delta P_0 = -\Delta c \rho g \quad (4.10)$$

Then, based on the obtained results, present the measured atmospheric pressure as follows:

$$P_0 \pm \Delta P_0 = (.. \pm ..) \quad (4.11)$$

You may use the linier regression method to determine the intercept and slope, along with their uncertainties, of the curve on the graph of h versus $1/l$. From the obtained slope and intercept, you can draw a straight line through the data points plotted on graph paper, which will resemble the graph shown in Figure 4.2.

4.4.2 Procedure

1. Turn the valve (S) counterclockwise so that the end of the left tube is open (see Figure 4.1). At this point, the mercury levels on both the left and right sides of the tube are equal. This occurs because both sides are exposed to the external air and thus experience equal pressure.
2. The distance between point S and the mercury surface on the left side is denoted by l , as shown Figure 4.1. The height of the right side of the U-tube can be

adjusted, while the left side remains fixed. When the height of the right side is changed, the value of l will also change. Adjust the height of the right tube so that l equals 20 cm . At this point, the mercury levels on both sides remain equal.

3. Turn the valve (S) clockwise so that the left end of the tube is closed.
4. When the right side of the manometer tube is raised, the value of l decreases. Adjust the height so that l becomes $19,5\text{ cm}$, then record the difference in the mercury level between the right and left sides. This height difference is denoted as h , as shown in fig 4.1.
5. Repeat step 4 for 10 variations of the height l with an interval of $0,5\text{ cm}$ (starting from $19,5\text{ cm}$, $19,0\text{ cm}$, $18,5\text{ cm}$, etc).
6. Plot the graph of h versus $1/l$ (see Figure 4.2).